

Yagyaansh Khaneja

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Education

IIT Roorkee, B. Tech. in Eng. Physics (CGPA 7.11/10) <i>First Division</i>	2021
RDPS, Delhi (CBSE), Class XII: PCM (94.2 %)	2017
RDPS, Delhi (CBSE), Class X (CGPA 10/10)	2015

Skills

- **Programming:** Python (PyTorch, NumPy, scikit-learn, pandas, matplotlib), FORTRAN, HTML/CSS, SQLite
- **Tools:** MATLAB, COMSOL, CST MW Studio, Excel/Word/PowerPoint, LaTeX, Lightroom
- **Languages:** English (Native), Hindi (Native), French (Basic), Punjabi (Basic)

Physics Projects

- **Research Intern (Energy Team) – Sonam Wangchuk** [Jan 2022 – May 2022]
 - **Organization:** Himalayan Institute of Alternatives ([HIAL](#)), Ladakh
 - **TL;DR:** Assisted renowned environmentalist [Sonam Wangchuk](#) on his projects (SDGs)
 - Drafted [manual](#) for construction of Passive Solar Heated ([PSH](#)) Buildings in Ladakh.
 - Energy Analysis of PCM in Tromb  Walls (PSH), and HVAC systems
 - Maintained and installed solar PV systems (20 kW, 7.5 kW)
 - Installed new Stevenson Screen; assisted in weather [data collection](#) (& analysis) in remote regions of Ladakh using SensorPush and Campbell Scientific weather stations.
 - **Skills:** Research (Thermal Physics), Excel, Project Implementation, Electrical Science
- **Bachelor’s Project – Silicon Photonic Switch, IIT Roorkee** [May 2020 – Sep 2021]
 - **Organization:** Physics Dept., IIT Roorkee ([Prof. Rajesh Kumar](#))
 - **TL;DR:** Simulated non-volatile GST-SOI waveguides using CST MW Studio and COMSOL.
 - Designed photonic switches for use at 2.1 m ([report](#)) and 1 THz ([thesis](#))
 - Studied Mux-DeMux using Optical Phased Array & WDM for telecom applications
 - **Skills:** EM Simulation ([COMSOL](#), CST MW Studio), MATLAB, Waveguide Photonics
- **Research Camp – Optics/Photonics, ITMO University** [May 2019 – Jun 2019]
 - **Organization:** ITMO University, St. Petersburg, Russia ([Prof. Anton Tcyppkin](#))
 - **TL;DR:** Gained exposure to advances in Optics through [active engagement](#) with frontier research ([Certificate](#))
 - Generated THz pulse radiation via laser filamentation of liquids ([First global attempt](#) using pH variation of water!)
 - Simulated interference & diffraction patterns for varying waveforms & apertures
 - Studied Quantum Information, Holography, QKD Systems, Femtosecond Lasers, etc.
 - **Skills:** Experimental Optics, MATLAB, Laser Systems

Achievements

- **NGPE 2019 (Part A):** Top 10%
- **JEE 2017 (Advanced):** AIR 3188
- **JEE 2017 (Mains):** AIR 2933 (Top **0.2%**)
- **Vijyoshi 2016 (Camp):** Invited Guest
- **KYPY 2015 (SA):** AIR 300 (Top **0.4%**)
- **NTSE 2015 (II):** Scholar (Top **0.1%**)

Certificates

- **Machine Learning:** Andrew Ng ([Coursera](#)) [2023]
- **Python for Everybody:** Charles Severance ([Coursera](#) Specialization) [2020]
- **Web Design for Everybody:** Colleen ven Lent ([Coursera](#) Specialization) [2020]
- **Internet: History, Technology, Security:** Charles Severance ([Coursera](#)) [2020]

Tech Projects

- **Vibe Coding Engineer** [Oct 2025 – present]
 - **Organization:** US based stealth StartUp (San Fransisco)
 - **TL;DR:** Built automation tools for the Hotel industry in the US
 - Wrote playwright scripts (Python) to scrape data from Hotel Management Systems (HMS) like ChoiceAdvantage, IHG Holiday Inn, etc.
 - Built a robust reconciliation engine to find discrepancies in data from different sources (HMS, CLC, Expedia, etc.)
 - **Results:** Got first 2 customers signed up within 1 week of prototype devt.
 - **Skills:** AWS deployment, Processing webpages, data (json, csv) in Python
- **Personal Website – yagyaansh.com** [Nov 2020 – Jan 2021]
 - **TL;DR:** Hard-coded my website using basic HTML and CSS (no frameworks)
 - Implemented Dark Theme toggle using JavaScript
 - Wrote blogs with graphics, integrated comments section using Disqus
 - Hosted my website using GitHub Pages
 - **Skills:** Web Dev (HTML, CSS, JS), handling SVG in HTML

Independent Study

- **CS50 (Stanford):** Intro to CS [2020]
- **CS50 (Stanford):** AI with Python [2020, 2021]
- **Andrej Karpathy:** NN: Zero to Hero [2024-25]
- **3Blue1Brown:** DL & Transformers [2025]
- **6.S191 (MIT):** Intro to Deep Learning [2025]
- **Kaggle:** Kaggle Learn [2025]

Side Projects

- **Standard Model - Visualized** [May 2025 – ongoing]
 - **TL;DR:** Visualized Standard Model of Particle Physics in 3D
 - Used three properties at a time (mass, charge, isospin/strangeness)
 - Tried matplotlib [initially](#), then switched to Manim for better [animations](#)
 - **Skills:** Python - matplotlib, Manim (by 3Blue1Brown)
- **Fee Data Extraction** [Aug 2020]
 - **TL;DR:** API calls (Azure server) to extract data of 3000+ students (college fee receipts)
 - Libraries used - urllib.request, ssl, json, sqlite3 in Python
 - converted the received json data into a SQLite database for easy accessibility
 - **Skills:** Handling JSON and SQLite database using Python

Multidisciplinary Projects

- **Family Business – B2B Wholesale (pan India)** [Sep 2022 – Sep 2025]
 - Location: Asia's largest wholesale market: Chandni Chowk/Chawri Bazaar (Delhi-6)
 - TL;DR: In essence, increased the efficiency of operations, **saving 4 hours per day**
 - Migrated **accounts** from legacy (handwritten) to cloud software, enabling remote work for the first time in 35 years
 - Meticulously classified 1000s of SKUs (**Steel products**), 400+ parties; making data entry organized
 - **Reverse engineered** API for a third-party app to automate data entry!
 - Cut operational overhead via warehouse optimization (Streamlined packaging, labels, item codes, colour coded dispatch, and process flows)
 - Expanded customer base through **online retail** sales via Meesho
 - Designed a catalog of Door Fittings, Fasteners, Screws - simplifying products for customers (Canva)
 - Learned invaluable lessons about Money, **Cash Flow**, Relationship Management, Business Ethics, Sales, etc.
 - Skills: Warehouse Management, Leadership, Product Design, Accounting & Financial Management
- **Leadership Roles, IIT Roorkee** [2017-2021]
 - Additional Secretary, Photography Section
 - Coordinator, Thomso (Cultural Fest)
 - Content Manager, Srishti (Tech Exhibition)
- Skills: Leadership & Management

Community Service

- **Project Manager – Tech Team** [Dec 2024 – Jun 2025]
 - Organizations: Tarayana, GIAOS
 - TL;DR: Guided the development of data dashboard (PoC)
 - Location: Wangdi, Bhutan
 - Empowered the remote village of Rukha in leveraging technology to advance the national vision of Gross National Happiness (GNH)
 - Under the aegis of renowned banker Tan Sri Andrew Sheng SBS
 - Skills: Team Management, High Level Architecture Design, (Tech Stack: React Native, MongoDB, DeepSeek API)
- **STEM & English Teacher**
 - Location: Ladakh (SECMOL) [2022]
 - Taught dropouts using hands-on science/maths experiments.
 - Led **astronomy** and photography hobby clubs.
 - Location: Vietnam (Workaway) [2022]
 - Taught English, Maths to students in the sub-urbs of Hanoi
 - Took part in activities with Sustainable Devt. Community in DaNang
 - Location: NSS, IIT Roorkee [2017-2018]
 - Taught English, Science & Maths to children from underprivileged communities around Roorkee
 - Skills: Communication, Empathy

Coursework (B. Tech.)

- **Basics**
 - **EMT**: Sadiku; Griffiths
 - **Relativity**: Resnick; Goldstein; Aurther Beiser
 - **QM 1**: Griffiths; Zettili
- **Maths**
 - **Mathematical Physics**: Mary L. Boas; Arfken
 - **Advanced Maths**: Kreyszig; Brown & Churchill
 - **Computational Physics**: DeVries (FORTRAN)
- **Nuclear**
 - **Nuclear Physics**: Krane; John Lilley
 - **Nuclear Astrophysics**: Rolfs & Rodney; Iliadis
- **Interdisciplinary**
 - **Thermal & Statistical Physics**: Sears & Salinger; Zemansky & Dittman; Landau & Lifshitz
 - **Atomic & Molecular Physics**: Haken & Wolf; Bransden
 - **Condensed Matter Physics**: Wahab
 - **Plasma Physics**: Francis F. Chen
- **Electronics**
 - **Electronics**: Malvino; Millman; Streetman; Horowitz; Scherz
 - **Electrical**: Chapman; Hughes
 - **Electrical & Electronic Materials**: S. O. Kasap
 - **Microprocessors**: Ramesh S. Gaonkar
 - **Semiconductor Devices**: Sze & Lee; Streetman
- **Optics/Photonics**
 - **Applied Optics**: Hecht; Born & Wolf; Ghatak
 - **Lasers & Photonics**: Ghatak & Thyagarajan; Silfvast; Svelto; Yariv
 - **Optical Communication**: G. P. Agrawal; Wartak; B. P. Lathi; Schaum
 - **Fiber & Non-Linear Optics**: Ghatak & Thyagarajan
 - **Optoelectronics**: Saleh & Teich; Kasap
 - **Photonic Sensors (Plasmonics)** : Sachin Srivastava; Maeir